AHMAD IBRAHIM TOLBA KHEDR

GIS & Stormwater Network Specialist | Hydrological Draftsman | InfraWizard Specialist | GOPP Expert Assistant | Infrastructure GIS

Location: Giza, Egypt | Phone: +20 111 191 5925 / +20 107 061 9064 | Email: ahmadibrahim.geo@gmail.com

LinkedIn: linkedin.com/in/ahmadibrahim4geo | GitHub: https://github.com/ahmadibrahim4geo

PROFESSIONAL SUMMARY

GIS and stormwater network specialist with 5+ years of experience in infrastructure GIS, CAD-to-GIS conversion, InfraWizard-based stormwater and groundwater reduction network preparation, dewatering systems, hydrological drafting support, and engineering deliverables. Experienced in building drainage and groundwater networks from survey data, SewerGEMS/StormCAD outputs, CAD drawings, and GIS datasets, with deliverables including PDF layouts, CAD drawings, GIS geodatabases/shapefiles, KML files, profiles, summaries, and surface/DEM outputs. Registered as a GOPP Expert Assistant, with practical project experience in Egypt and Saudi Arabia, especially Jeddah stormwater, groundwater control, urgent drainage solutions, and alternative drainage routes.

SPECIALIZED EXPERTISE

InfraWizard Stormwater & Groundwater Networks

Manual network building, structure/pipe setup, profile extraction, network coloring, layout preparation, and practical handling of InfraWizard limitations.

SewerGEMS / StormCAD Data Preparation

Extracting and cleaning FlexTable outputs, preparing shapefiles, separating pipe diameter and material fields, and resolving data formats before InfraWizard import.

GIS / CAD / KML Deliverables

Preparing geodatabases, shapefiles, CAD drawings, KML files, PDF layouts, network summaries, and map outputs for engineering teams and clients.

Survey-to-Network Workflows

Converting field survey levels, coordinates, pipe diameters, materials, ground levels, invert levels, and depths into structured drainage network models.

Network Review & QA/QC Support

Checking negative slopes, invert/ground level consistency, pipe sizes, naming conventions, depth values, protection requirements, and engineering presentation quality.

Surface & Raster Preparation

Generating road/survey-based surfaces from points and centerline/edge data, preparing TIF raster surfaces for use in Civil 3D, SewerGEMS, StormCAD, ICM, and related modeling workflows.

CORE TECHNICAL SKILLS

- **GIS & Spatial Analysis:** ArcGIS, QGIS, geodatabase design, spatial analysis, map production, terrain/DEM analysis, shapefile/KML handling.
- **Stormwater & Groundwater:** Stormwater drainage networks, groundwater reduction networks, dewatering systems, urgent drainage solutions, alternative drainage routes, hydrological report support.
- **InfraWizard & Engineering Data:** InfraWizard network preparation, SewerGEMS/StormCAD data preparation, FlexTable cleanup, pipe/manhole attributes, profiles, summaries, layouts.
- **CAD & Civil 3D Support:** AutoCAD 2D, Civil 3D layout support, CAD-to-GIS conversion, utility networks, engineering plans, profiles, basic surface extraction.
- **Data, Automation & Documentation:** Excel, data cleaning, technical documentation, workflow improvement, AI-assisted GIS problem solving and automation concepts.

• **Geospatial Programming & GIS Automation:** Python and ArcPy toolboxes for ArcMap / ArcGIS Pro; geoprocessing, climate-data workflows, cartographic automation, spatial data validation, and raster/vector outputs.

• **Desktop, Mobile & Web Software Development:** C# / WPF / .NET desktop utilities, Kotlin / Android GIS applications, HTML / CSS / JavaScript dashboards, Git/GitHub version control, and bilingual software documentation.

SOFTWARE / TOOLS & LANGUAGES

- **Software / Tools**: InfraWizard, ArcMap ArcGIS Desktop, ArcGIS Pro, QGIS, Global Mapper, AutoCAD, Civil 3D, SewerGEMS / StormCAD data preparation, SAS Planet, Microsoft Office Suite, Google Earth Pro, Google Earth Engine, Payazed Menu.

  Additional Technical Skills: Adobe Illustrator, Photoshop, and InDesign.
- **Languages**: Arabic: Native | English: Basic - able to read technical GIS/CAD terms and use software interfaces with support from translation tools.

• **Software Development Technologies:** AI-assisted programming using Python / ArcPy, C# / .NET / WPF, Kotlin / Jetpack Compose, JavaScript / HTML / CSS, Git/GitHub, desktop installers, and GIS data services.

SOFTWARE TOOLS DEVELOPMENT (2026)

Developed and published GIS, climate, geology and hydrology projects on GitHub, followed by general desktop, mobile and web applications. Selected repositories and current development work are listed below.

GIS, CLIMATE, GEOLOGY & HYDROLOGY SOFTWARE (2026)

• **Atlas of Egyptian Curriculums Maps App:** Developed a maps application for Egyptian curriculum atlas and map workflows.

• **MIRSAD - Climate Atlas Platform:** Developed ArcMap / ArcGIS Pro Python toolboxes for weather-station selection, climate indicators, spatial interpolation and atlas outputs; includes 19 climate modules and a bilingual standalone Windows desktop development build. | GitHub

• **CreateFlowArrow - Gravity Network QA/QC Toolbox:** Developed an ArcPy toolbox for stormwater, sewer and dewatering networks, generating flow arrows, calculating invert-based slopes, flagging direction / connectivity defects, and optionally correcting reversed pipes. | GitHub

• **Geological Cross-Section Toolbox for ArcGIS:** Developed automated 2D geological and topographical sections from outcrops, faults and local / online elevation data, with batch processing and geodatabase, DXF, HTML and tabular outputs. | GitHub

• **GeoVision - Mobile GIS Viewer:** Developed an Android GIS viewer in Kotlin / Jetpack Compose for local vector and raster datasets, including Shapefile, GeoJSON, KML, GeoPackage and GeoTIFF, with Arabic / English interfaces and offline data workflows. | GitHub

• **Storm Drainage Calculation Dashboard:** Developed a browser-based tool for preliminary stormwater calculations: IDF relationships, SCS Curve Number runoff, design rainfall distributions and Rational Method peak discharge, with charts, CSV export and printable reports. | GitHub

• **GIS Style Collection:** Curated and published ArcMap (.style) and ArcGIS Pro (.stylx) cartographic resources across 27 categories, with bilingual search, Arabic style variants and translation scripts; covers geology, hydrology, soils, land use and infrastructure. | GitHub

• **GIS Data Repositories Professional 2026:** Built a curated catalog of 2,660+ GIS data sources with a bilingual searchable portal, category filters, source-link audits, a map explorer, and Excel / CSV / JSON exports for geospatial data discovery. | GitHub

• **GIS Development Skills:** Authored a bilingual skill suite for AI-assisted GIS development across ArcGIS, QGIS, desktop and web mapping, spatial data pipelines and API integrations, with specialist routing, reproducible examples and evidence-based verification workflows. | GitHub

CURRENT DEVELOPMENT / IN-PROGRESS APPLICATIONS

• **Z EARTH - Desktop GIS & 3D Globe:** Developing a Windows GIS application using C# / WPF, CesiumJS / WebView2 and GDAL / PROJ for KML/KMZ, Places / Layers and 2D/3D mapping. Source and portable preview are available; stabilization remains in progress. | GitHub

OTHER DEVELOPED DESKTOP, MOBILE & WEB APPLICATIONS (2026)

• **SmartCalculator:** Developed a C# / WPF scientific calculator with complex numbers, statistics, equation solving, matrices, vectors, number-base conversion and function tables; includes compact mode and persistent preferences. | GitHub

• **SmartPhotoOrganizer:** Developed a Windows WPF / .NET photo-management application for EXIF / GPS review, duplicate detection, image-quality assessment, bulk renaming, map-based photo browsing and CSV / PDF reporting. | GitHub

• **RoboTransfer:** Developed a bilingual Windows / .NET interface for Robocopy with copy, update, mirror and move modes, dry-run previews, live progress, transfer summaries and Unicode logs. | GitHub

• **DropAndCopy:** Developed a Windows drag-and-drop file collection and transfer utility with copy / move workflows, automatic conflict renaming, optional folder-structure preservation and progress tracking. | GitHub

• **Charge Alert (Android):** Developed a Kotlin / Jetpack Compose battery-monitoring application with configurable charge-target alerts, background monitoring, Arabic / English interfaces and selectable charging designs. | GitHub

• **FitraShield - Browser Safety Extension:** Developed a JavaScript browser extension with local content filtering, custom block / allow lists, parental reporting and a Python / Windows helper for browser-policy integration. | GitHub

• **Ahmad Ibrahim Portfolio & Interactive CV:** Developed a responsive English / Arabic portfolio and interactive CV presenting infrastructure projects, registered atlases, publications, professional capabilities and software repositories. | GitHub

PROFESSIONAL EXPERIENCE

GIS Specialist & Hydrological Drafting Specialist | Vision Dimensions Engineering Consultancy Oct 2021 - Dec 2024

- Built and prepared stormwater drainage and groundwater reduction networks using InfraWizard based on survey data, pipe diameters, materials, ground levels, invert levels, and engineering constraints.
- Converted CAD drawings, field survey outputs, and engineering model exports into structured GIS datasets, shapefiles, geodatabases, KML files, CAD outputs, and PDF layouts.
- Prepared InfraWizard-ready data from SewerGEMS/StormCAD outputs, including FlexTable extraction, field cleanup, pipe size/material separation, and shapefile processing before import.
- Reviewed stormwater and groundwater networks for negative slopes, depth/invert consistency, pipe diameter suitability, naming quality, protection requirements, and presentation standards.
- Prepared engineering plans, profiles, network summaries, viewports, layouts, and technical mapping outputs for urgent drainage solutions, alternative routes, and groundwater control projects.
- Generated survey/road-based surfaces and raster TIF outputs from point data and road centerline/edge levels to support hydraulic modeling and Civil 3D/GIS workflows.
- Supported hydrological report preparation through data collection, spatial analysis, map production, technical documentation, and workflow improvement.

Freelance GIS Specialist | Independent Projects Jan 2020 - Oct 2021

- Delivered GIS and mapping support for infrastructure network projects including electricity, gas, sewage, and utility networks.
- Prepared maps and spatial outputs for master's and doctoral theses, academic research, and registered geographical atlases.
- Conducted GIS training sessions and supported learners in practical map production, terrain analysis, and remote sensing workflows.

SELECTED PROJECT EXPERIENCE

- **Stormwater & Groundwater Network Projects - Jeddah, KSA:** Supported and prepared dozens of urgent drainage solutions, alternative drainage routes, groundwater reduction lines, and stormwater layouts using InfraWizard, GIS, CAD, KML, and PDF deliverables.
- **AL Jawaharah Mall Stormwater Drainage Network - Jeddah, KSA:** Prepared stormwater drainage layout support including manholes, catch basins, pipe diameters, slopes, invert levels, depths, coordinates, overflow pipes, and network summary outputs.
- **Groundwater Network Survey Project - Jeddah, KSA (2022):** Developed and managed GIS/CAD datasets for groundwater and drainage networks, including survey data conversion, plan extraction, and engineering profile preparation.
- **Stormwater Drainage Networks - Jeddah Municipality (2023):** Contributed to network drafting and GIS support across multiple planning zones, including Plans 583/G/S, 654/G/S, 677/G/S, 679/G/S, 680/G/S, 681/G/S, 682/G/S, 683/G/S, 704/G/S, 705/G/S, and 718/G/S.
- **Groundwater Control & Dewatering Systems - Jeddah, KSA:** Prepared drafting and GIS support for commercial, residential, and infrastructure dewatering projects in Alruwais, Alhamraa, Alkhalidiah, Alaziziah, Alasalah, and other districts.
- **Flood Protection for Wadi Laban - Riyadh, KSA:** Supported mapping, drafting, and engineering documentation related to flood protection works.
- **CAD-to-GIS Electrical Plans - Jeddah, KSA:** Converted electrical plans from CAD files into GIS databases, including Plans 675, 679, and 680.
- **Climate Resilient ICZM Plan for the North Coast of Egypt (2023):** Contributed GIS and mapping support to climate-resilient integrated coastal zone management documentation.

PUBLICATIONS & SCIENTIFIC CONTRIBUTIONS

- Authored **Atlas of Egypt, the Arab World and the World**, registered with the National Library and Archives of Egypt under Local Deposit No. 2021/20535 and ISBN 9789779095615.
- Authored **Atlas of the World and the Arab World**, registered with the National Library and Archives of Egypt under Local Deposit No. 2022/15361 and ISBN 9789779425962.
- Authored **Atlas of Geography of Egypt**, registered with the National Library and Archives of Egypt under Local Deposit No. 2022/15362 and ISBN 9789779425979.
- Authored **Atlas of Geography of Development**, registered with the National Library and Archives of Egypt under Local Deposit No. 2022/15363 and ISBN 9789779425986.
- Authored **Atlas of Political Geography**, registered with the National Library and Archives of Egypt under Local Deposit No. 2022/15364 and ISBN 9789779425993.
- Authored **Atlas of Blank Maps**, registered with the National Library and Archives of Egypt under Local Deposit No. 2022/15365 and ISBN 9789779426006.
- Reviewed **the 2020 edition of the World Atlas**, produced by Dar Al-Irshad, Cairo Branch.
- **Participated in producing the Atlas of the Nile Basin Countries - Geological Aspect** at the Maps and GIS Division, Department of Geography, Faculty of Arts, Cairo University, 2018.
- Prepared maps for several scientific research papers, master's theses, and doctoral dissertations.
- One year of experience delivering GIS and remote sensing training sessions.
- Delivered online training sessions via Zoom in 2021 on GIS applications, including slope aspect mapping, surface roughness mapping, local elevation mapping, and alternative terrain shadow mapping techniques.

EDUCATION

Bachelor of Arts in Geography, Maps & Geographic Information Systems Specialization | Faculty of Arts, Cairo University | May 2018

CERTIFICATES AND TRAINING COURSES

Esri

- **ArcGIS Online Basics** | October 16, 2025.
- **Cartography** | October 11, 2025; also completed October 13, 2018 (6 weeks each).
- **Working with Geodatabase Domains and Subtypes in ArcGIS** | August 25, 2025.
- **Getting Started with the Geodatabase** | August 25, 2025 | 1 hour 55 minutes.
- **Georeferencing Raster Data Using ArcGIS** | July 28, 2025.
- **Introduction to Coordinate Systems** | July 26, 2025.
- **GIS Basics** | December 21, 2023.
- **ArcGIS Maps for Office Basics** | December 13, 2019.
- **Getting Started with GIS** | September 18, 2018.

Udemy

- **Model Builder ArcGIS** | February 11, 2020 | 13.5 hours.
- **Presentation Skills (in Arabic)** | August 16, 2018.

GEO University

- **Digital Image Processing with OpenCV in Python** | September 17, 2018.
- **Hyperspectral Thermal Image Unmixing** | September 17, 2018.

University of California, Davis via Coursera

- **Fundamentals of GIS** | August 31, 2018.

Ain Shams University

- **Geomatics** | July 10 - August 17, 2017 | Center for Studies, Consultations and Training, Faculty of Arts.

National Authority for Remote Sensing and Space Sciences (NARSS)

- **Remote Sensing and GIS Summer Training** | August 21-25, 2016 | Remote sensing, image analysis, ENVI and GIS.

African Center for Surveying Services (ACSS)

- **Applied Surveying** | August 2016 | 60 hours | Level, Total Station, AutoCAD and Civil 3D.

Additional Training

- **Planning and Geographic Information Systems Course** | May 1 - August 30, 2024.